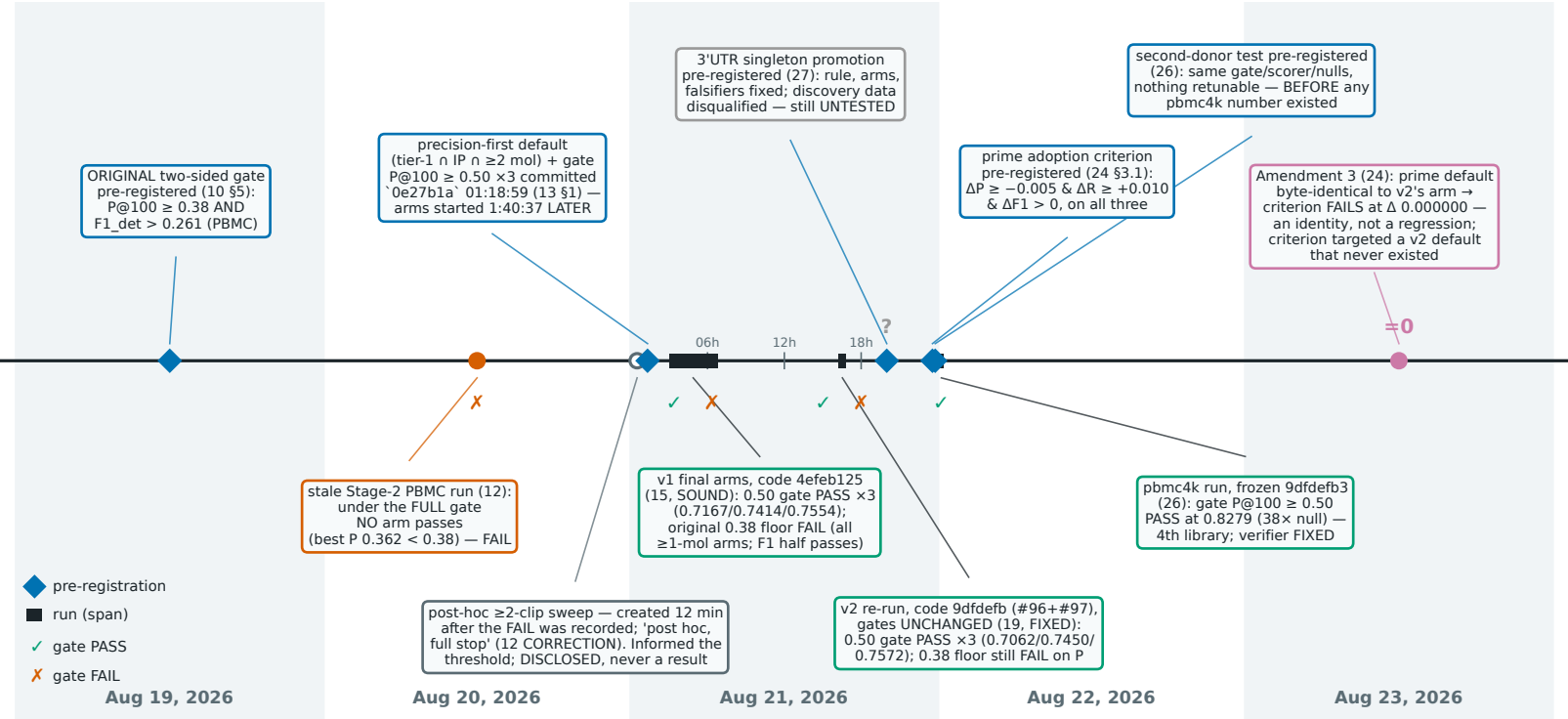


a The pre-registration timeline — every gate fixed before the number it judges



b Every pre-registered gate and every outcome (table T3 drawn)

ORIGINAL two-sided gate — $P@100 \geq 0.38$ AND $F1_det > 0.261$, PBMC ≥ 1 -molecule output (10 \$5, 2026-08-19)		
Stage-2 run 2026-08-20 (stale; 12)	both tiers n 377,236: P 0.188 / F1 0.235; tier-1 n 197,410: P 0.308 / F1 0.290; tier-1+IP n 142,400: P 0.362 / F1 0.290 — 'under the full gate, NO arm passes'	FAIL
v1 final run 2026-08-21 (15 \$2)	tier-1 ≥ 1 mol no-IP n 222,955: P 0.3032 / F1 0.3001; IP arm n 161,595: P 0.3546 / F1 0.3013	F1 pass / P FAIL
v2 re-run 2026-08-21 (19 \$2)	tier-1 ≥ 1 mol no-IP n 222,955: P 0.3032 (byte-identical to v1); IP arm n 167,565: P 0.3520 / F1 0.3046	F1 pass / P FAIL
PRECISION-FIRST default gate — $P@100 \geq 0.50$ on PBMC AND both mice (13 \$1; commit '0e27b1a' 01:18:59, 1:40:37 before the arms)		
v1 final run 2026-08-21 (15 \$1)	0.7167 / 0.7414 / 0.7554 (n 44,394 / 25,991 / 26,164)	PASS $\times 3$
v2 re-run 2026-08-21, gates unchanged (19 \$1)	0.7062 / 0.7450 / 0.7572 (n 46,524 / 26,255 / 26,526)	PASS $\times 3$
pbmc4k second donor 2026-08-22 (26 \$6, same gate)	0.8279 (n 20,672; 38 $\times$ its genic-shuffle null) — the 4th library	PASS
PRIME adoption criterion — $\Delta P \geq -0.005$ & $\Delta R \geq +0.010$ & $\Delta F1_det > 0$ on all three datasets, default vs default (24 \$3.1, 2026-08-21; all prime numbers EXPLORATORY per \$3.4)		
prime default vs v2 default (Amendment 3, 2026-08-23)	prime's default output byte-identical to v2's manuscript arm on all four libraries → every Δ exactly 0.000000 — an identity, not a regression; the criterion was written against a v2 default that never existed (—ip-filter-mode default is 'annotate')	FAIL (identity)
3'UTR SINGLETON promotion — R2 arm; no precision loss with a recall gain, deciding evidence = testis mouse 2 + non-discovery data (27 \$1/\$3, 2026-08-21; Rule T)		
confirmatory evaluation	NOT RUN — discovery-set numbers are labelled 'DISCOVERY SET — not evidence'; the promotion ships behind a flag, default OFF, until 27 \$3 passes	PENDING

Fig S11 | The pre-registration record, drawn so it can be audited rather than taken on trust. (a) Timeline. The original two-sided gate ($P@100 \geq 0.38$ AND $F1_det > 0.261$) was fixed in 10 \$5 on 2026-08-19, one day before the run it judged; the stale Stage-2 run FAILED it in full (12: 'NO arm passes'). The ≥ 2 -molecule threshold was then examined in a sweep created 12 minutes after that FAIL was recorded — post hoc, full stop (12 CORRECTION) — and is disclosed as such: it informed the choice and is never cited as a result. The precision-first default and its $P@100 \geq 0.50$ gate were committed ('0e27b1a', 01:18:59, git-verified) 1:40:37 before the final arms started (02:59:36); the v1 run passed on all three call sets, and the v2 re-run (code 9dfdefb, 16:14, gates UNCHANGED) passed again while the original 0.38 precision floor kept failing on every ≥ 1 -molecule output — both facts shown, neither hidden. Three further pre-registrations followed on 2026-08-21, each written before its number existed: the prime adoption criterion (24, last write 23:34), the second-donor test (26, 23:47, driver frozen 19 min before the caller started), and the 3'UTR singleton promotion (27, still untested). Outcomes: pbmc4k PASS 0.8279 (2026-08-22, verifier FIXED); prime FAIL at $\Delta 0.000000$ — a byte-identity, not a regression, because the criterion was drafted against a v2 default that never existed (Amendment 3, 2026-08-23); singleton PENDING, flag default OFF. Markers whose documents record a date but no clock time sit at display positions, flagged in the audit TSV. (b) Table T3 drawn: every pre-registered gate, every arm it was evaluated on, every value and every verdict — including both FAILs and the untested pre-registration. Gate values are re-read from the score TSVs where they exist (v1 arms, v2 via fig2_accuracy.tsv, pbmc4k) and EXPECT-asserted against 15/19/26; Stage-2 values are quoted from 12 (its matrices were voided; its own verifier reproduced them to 6 decimals). Every event and row: results/figures/manuscript/figS11_gatehistory*.tsv.